

1. Administrative & Study Identification

Official Title: A Phase III, Randomized, Double-Blind, Placebo-Controlled Study Evaluating the Efficacy and Safety of Inavolisib Plus Palbociclib and Fulvestrant Versus Placebo Plus Palbociclib and Fulvestrant in Patients With PIK3CA-Mutant, Hormone Receptor-Positive, HER2-Negative, Locally Advanced or Metastatic Breast Cancer **Brief Title:** A Study Evaluating the Efficacy and Safety of Inavolisib + Palbociclib + Fulvestrant vs Placebo + Palbociclib + Fulvestrant in Participants With PIK3CA-Mutant, Hormone Receptor-Positive, HER2-Negative, Locally Advanced or Metastatic Breast Cancer **Short Title/Acronym:** INAV0120 **Protocol Number:** W041554 **NCT Number:** NCT04191499 **Posting ID:** 188928349158633105 **Sponsor/Company Name:** Hoffmann-La Roche / Roche **Investigational Product:** Inavolisib (Itovebi) **Phase of Development:** Phase II / Phase III **Trial Status:** Active, Not Recruiting **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of inavolisib in combination with palbociclib and fulvestrant compared with placebo plus palbociclib and fulvestrant, as measured by investigator-assessed Progression-Free Survival (PFS). **Secondary Objectives:** To evaluate Overall Survival (OS), Objective Response Rate (ORR), Duration of Response (DOR), Best Overall Response Rate (BOR), Clinical Benefit Rate (CBR), safety and tolerability, and patient-reported outcomes.

2.2 Background & Rationale To address the need for targeted therapies in patients with PIK3CA-mutated, HR+/HER2- locally advanced or metastatic breast cancer with endocrine resistance. PI3K, CDK4/6, and Estrogen Receptor (ER) pathways are highly interconnected and drive endocrine resistance. Inavolisib is a highly potent, selective PI3K inhibitor that promotes the degradation of mutated p110α. Combining it with a CDK4/6 inhibitor (palbociclib) and an ER degrader (fulvestrant) aims to delay disease progression and improve survival outcomes for a patient population with an historically poor prognosis.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized (1:1)

3.2 Planned Enrollment & Duration Targeted Enrollment: 325 participants **Number of Sites:** Multicenter, global study **Estimated Study Start:** 29-Jan-2020 **Estimated Primary Completion:** November 2024 (Data Cutoff for OS Analysis)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years, male or female. 2. Locally advanced or metastatic Hormone Receptor-positive (HR+), HER2-negative breast cancer with a qualifying PIK3CA mutation (detected via FDA-approved ctDNA or tissue test). 3. Disease relapsed during or within 12 months after completing adjuvant endocrine therapy with an aromatase inhibitor or tamoxifen. 4. Fasting glucose < 126 mg/dL and HbA1c $< 6.0\%$.

4.2 Exclusion Criteria 1. Prior systemic therapy for metastatic disease. 2. Prior treatment with a PI3K, AKT, or mTOR inhibitor. 3. Type 1 diabetes mellitus, or Type 2 diabetes mellitus requiring ongoing antihyperglycemic treatment at the start of study treatment. 4. Known and untreated, or active central nervous system (CNS) metastases (previously treated CNS metastases allowed).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Inavolisib 9 mg once daily + Palbociclib (standard 21 days on, 7 days off) + Fulvestrant (Trade Name: Faslodex | ATC Code: L02BA03) (standard dosing schedule on Days 1 and 15 of cycle 1, then every 4 weeks). **Route of Administration:** Oral (inavolisib, palbociclib) and Intramuscular Injection (fulvestrant). **Duration of Treatment:** Administered in 28-day cycles until disease progression, unacceptable toxicity, or withdrawal of consent.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care; Pre/perimenopausal women and men must receive concurrent luteinizing hormone-releasing hormone (LHRH) agonist therapy for ovarian/testicular suppression. **Prohibited:** Other anticancer therapies; live attenuated vaccines; strong CYP3A inhibitors/inducers where applicable.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, PIK3CA Mutation Testing, Fasting Glucose/HbA1c Labs
Baseline	Day 0	Randomization (1:1), First Dose of Study Treatment
Interim	Weekly (Cycle 1) then Day 1 of each 28-day cycle	Safety Labs (FPG/FBG monitoring), Efficacy/Radiological Assessments, AE monitoring
End of Study		

Follow-up Phase | 1 month post-treatment safety visit, then survival follow-up every 2-3 months |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Investigator-assessed Progression-Free Survival (PFS); defined as the time from randomization to the first occurrence of disease progression or death from any cause. **Measurement Method:** Radiological assessment per Response Evaluation Criteria in Solid Tumors (RECIST) version 1.1.

7.2 Secondary & Exploratory Endpoints 1. Overall Survival (OS) 2. Objective Response Rate (ORR) and Best Overall Response Rate (BOR) 3. Duration of Response (DOR) 4. Clinical Benefit Rate (CBR)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via onsite visits and lab monitoring, strongly focused on PI3K inhibitor-related toxicities (hyperglycemia, stomatitis, diarrhea) and palbociclib-related neutropenia. AEs graded per NCI-CTCAE v5.0.

Serious Adverse Events (SAEs): Routine reporting timeline, including specific tracking for ketoacidosis and severe hypersensitivity. **Data Monitoring Committee (DMC):** Independent external DMC responsible for periodic safety oversight and review.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for all primary and secondary efficacy endpoints; Safety Set for AE/SAE evaluations. **Sample Size Justification:** Target sample size of $N=325$ to achieve appropriate power (approx. 85%) to detect a clinically meaningful improvement in PFS. **Handling of Missing Data:** Censoring rules applied for time-to-event outcomes (PFS/OS) utilizing Kaplan-Meier methodologies and stratified Cox proportional hazards models.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Scheduled onsite and remote clinical monitoring visits by the sponsor to ensure protocol adherence. **Audit Trail:** Validated eCRF locking mechanisms, centralized laboratory test tracking, and formal data clarification protocols.

11. Study Identifiers & Governance

Primary ID: W041554 **Registry Number:** NCT04191499 **Posting ID:** 188928349158633105 **Sponsor/Lead Organization:** Roche **Study Title:** INAV0120 **Trial Status:** Active, Not Recruiting

12. Clinical Framework (Breast Cancer Specific)

Disease Area: Breast Cancer **Preferred Disease Name:** Malignant neoplasm of breast **Preferred UMLS Name:** Estrogen receptor positive, HER2 negative breast cancer **MeSH Code:** D001943 **SNOMED ID:** 254837009 **Primary Condition:** PIK3CA-mutated, HR-positive, HER2-negative advanced or metastatic breast cancer **Diagnosis Threshold:** PIK3CA mutation positive by FDA-approved test (ctDNA or tumor tissue) **Medical Methodology:** PI3K inhibitor + CDK4/6 inhibitor + Endocrine therapy (ER degrader) combination therapy **Optimization Target:** Significant delay in disease progression (PFS) and extension of Overall Survival (OS)

13. Study Procedures & Methodology

Management Pathway: Standard oncology clinical pathway for advanced breast cancer **Education Protocol (HCP):** Dose modification and toxicity management training (specifically targeting hyperglycemia, neutropenia, and stomatitis) **Education Protocol (Patient):** Strict guidance on fasting blood glucose monitoring, recognizing signs of ketoacidosis, and oral drug administration routines **Quality Audit Mechanism:** Centralized laboratory biomarker confirmation and independent review facility/investigator tumor assessment cross-checks.

14. Observation & Follow-Up Schedule

Total Duration: Median follow-up $\geq$ 34 months for long-term Overall Survival evaluation **Onsite Visit Cadence:** Weekly during Cycle 1, then Day 1 of every 28-day cycle **Remote Monitoring Frequency:** Telephone calls/remote checks permitted; survival follow-up every 2 to 3 months post-treatment **Checklist Review Interval:** Routine Q4W (every cycle) lab and vital check

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				